

1 Solution — GIAM 4.1

Problem 7

1.1 Problem

Find a logical open sentence such that $\{0, 1, 4, 9, \dots\}$ is its truth set.

1.2 Answer

$$P(x) = \text{“}\exists y \in \mathbb{N} \text{ such that } x = y^2\text{”}$$

in words, “ x is a perfect square”. Working in the universe $\mathbb{N}$,

$$S_P = \{x \in \mathbb{N} : \exists y \in \mathbb{N}, x = y^2\} = \{0, 1, 4, 9, 16, 25, \dots\}.$$

1.3 Reading the Pattern

The listed elements are

$$0 = 0^2, \quad 1 = 1^2, \quad 4 = 2^2, \quad 9 = 3^2,$$

so the intended set is the perfect squares. The ellipsis has to be interpreted, and this is the reading it invites — the next entries would be **16, 25, 36**, and the truth set of P above is exactly that, verified for every $x < 400$.

An equivalent phrasing avoiding the quantifier is

$$Q(x) = \text{“}\sqrt{x} \text{ is an integer”},$$

which has the same truth set — also verified. Either answers the question.

1.4 The Picture

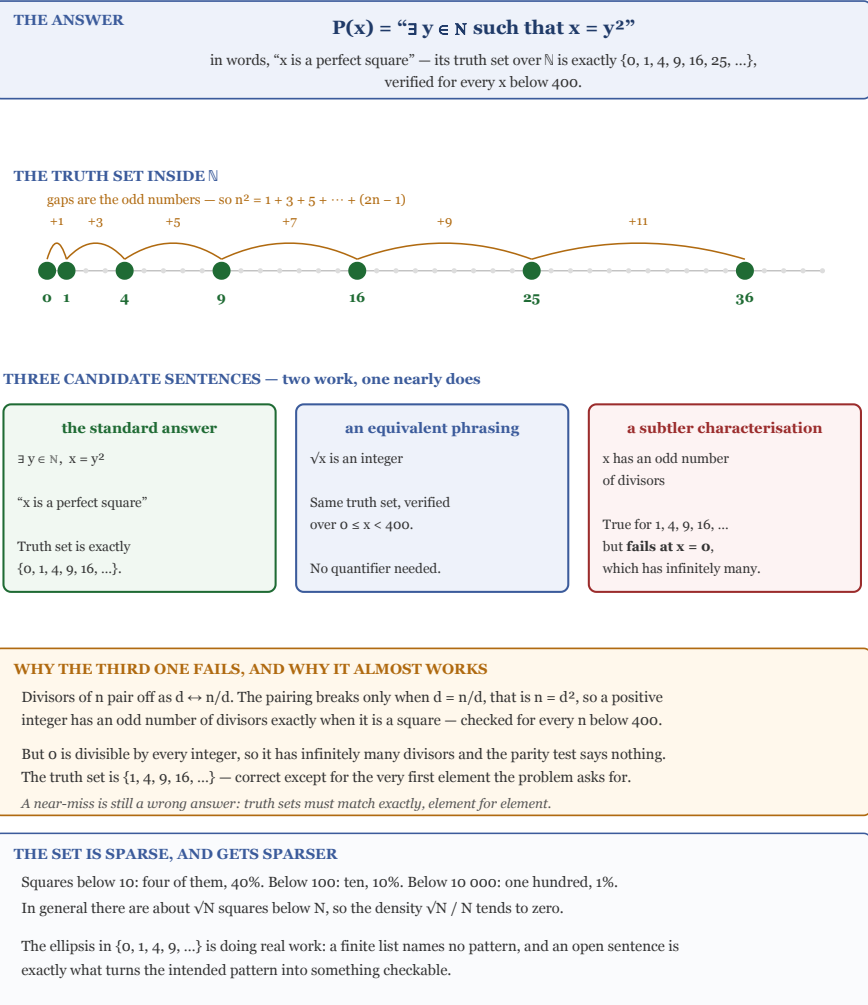


Figure 1.1: A diagram in four parts. The first gives the answer: P of x is the sentence there exists a natural y such that x equals y squared, in words x is a perfect square; its truth set over the naturals is exactly 0, 1, 4, 9, 16, 25 and onward, verified for every x below 400. The second part draws the naturals from 0 to 40 along a line, with the perfect squares 0, 1, 4, 9, 16, 25 and 36 drawn as large filled green circles and labelled, and every other integer as a faint grey dot. Amber arcs join consecutive squares and are labelled with the gaps: plus 1, plus 3, plus 5, plus 7, plus 9, plus 11. A note reads that the gaps are the odd numbers, so n squared equals 1 plus 3 plus 5 and so on up to $2n$ minus 1. Three candidate sentences follow as cards. The first, in green, headed the standard answer: there exists y in the naturals with x equal to y squared, meaning x is a perfect square, whose truth set is exactly 0, 1, 4, 9, 16 and onward. The second, in blue, headed an equivalent phrasing: the square root of x is an integer, the same truth set verified over 0 to 400, needing no quantifier. The third, in red, headed a subtler characterisation: x has an odd number of divisors, true for 1, 4, 9, 16 and so on but failing at x equal to 0, which has infinitely many divisors. A band below explains why the third fails and why it almost works: divisors of n pair off as d with n over d , and the

pairing breaks only when d equals n over d , that is n equals d squared, so a positive integer has an odd number of divisors exactly when it is a square, checked for every n below 400. But 0 is divisible by every integer, so it has infinitely many divisors and the parity test says nothing; the truth set is 1, 4, 9, 16 and onward, correct except for the very first element the problem asks for. A near-miss is still a wrong answer, since truth sets must match exactly, element for element. A final band notes the set is sparse and gets sparser: four squares below 10, or 40 percent; ten below 100, or 10 percent; one hundred below 10000, or 1 percent. In general there are about the square root of N squares below N , so the density tends to zero. The ellipsis in the listed set is doing real work, since a finite list names no pattern and an open sentence is exactly what turns the intended pattern into something checkable.

1.5 Remark — The Exercise Runs the Dictionary Backwards

Problems 4 through 6 went from a sentence to its truth set. This one goes the other way: a set is given and a sentence must be produced.

$$\begin{array}{l|l} \text{Problems 4–6} & P(x) \mapsto S_P \\ \text{this problem} & S \mapsto P \text{ with } S_P = S \end{array}$$

The backwards direction is genuinely harder, for two reasons. First, the set is presented by a finite list plus an ellipsis, so one must *guess the pattern* — and no finite list determines a pattern uniquely. Second, the answer is not unique even once the pattern is fixed, since many sentences share a truth set. That was the many-to-one observation from Problem 4, and here it works in one’s favour: any correct description will do.

It is worth noting that “ $x \in \{0, 1, 4, 9, \dots\}$ ” is *not* an acceptable answer, even though it is technically a sentence with the right truth set. It names the set rather than describing it, so it explains nothing about which numbers belong. The exercise wants a property, not a re-labelling.

1.6 Remark — A Third Characterisation, and Why It Just Misses

A more interesting description of the squares is available:

*A positive integer has an **odd number of divisors** exactly when it is a perfect square.*

The reason is a pairing argument. The divisors of n come in pairs $d \leftrightarrow n/d$, which exhausts them; a divisor is left unpaired only when $d = n/d$, that is when $n = d^2$. So the count is even unless n is a square.

n	divisors	count	square?
12	1, 2, 3, 4, 6, 12	6	no
16	1, 2, 4, 8, 16	5	yes
36	1, 2, 3, 4, 6, 9, 12, 18, 36	9	yes

Verified for every n below 400. And yet as an answer to this exercise it **fails**, because of 0: every integer divides 0, so 0 has infinitely many divisors and the parity test does not apply. The truth set is $\{1, 4, 9, 16, \dots\}$ — missing precisely the first element the problem lists.

That is worth dwelling on. A description that is right for all but one element is simply a wrong answer here: truth sets must agree element for element, and the boundary cases are where such descriptions usually break. The same caution applied in Problem 4, where “ $x > 20$ ” had the right truth set only relative to one particular universe.

1.7 Remark — The Universe Matters, and Here It Is Forgiving

Since a truth set depends on the universe of discourse, a complete answer should say what the universe is. The problem implies $\mathbb{N}$, and the given sentence works there.

Pleasantly, it survives a change: taking the universe to be $\mathbb{Z}$ and letting y range over $\mathbb{Z}$ gives the **same** truth set, since $y^2 = (-y)^2$ and no negative integer is a square. Confirmed over $-50 \leq x < 200$.

It does not survive every change, though. Widening the range of the quantified y eventually destroys the answer:

y ranges over	truth set within $\mathbb{N}$
$\mathbb{N}$ or $\mathbb{Z}$	the squares — correct
$\mathbb{Q}$	still the squares
$\mathbb{R}$	all of $\mathbb{N}$ — useless

The $\mathbb{Q}$ row is not obvious but is true: a rational whose square is a natural number must be an integer, and a search over all non-integer rationals with denominator up to 20 turns up no counterexample. The $\mathbb{R}$ row is the failure — every natural x equals $(\sqrt{x})^2$, so the sentence becomes vacuously true and its truth set is everything.

So both the universe of x and the range of y must be pinned down. The compact phrase “ x is a perfect square” quietly assumes both, which is fine in context but worth noticing.

1.8 Remark — The Squares Are Sparse, and the Gaps Are the Odd Numbers

Two structural facts about the answer set, both visible in the figure.

The gaps. Consecutive squares differ by consecutive odd numbers:

$$(n+1)^2 - n^2 = 2n+1,$$

so the list 0, 1, 4, 9, 16, 25 advances by 1, 3, 5, 7, 9. Summing gives the classical identity

$$n^2 = 1 + 3 + 5 + \cdots + (2n-1),$$

verified for $n < 50$. So an alternative open sentence is “ x is a sum of consecutive odd numbers starting from 1” — which does include 0, as the empty sum.

The density. Counting squares below N :

N	squares below N	density
10	4	40%
100	10	10%
1000	32	3.2%
10 000	100	1%

There are about $\sqrt{N}$ of them below N , so the density $\sqrt{N}/N = 1/\sqrt{N}$ tends to 0. The truth set is infinite but vanishingly thin — which is exactly why a defining property is more useful than any amount of listing.